DATA MINING PROJECT: The Smart-City Traffic Analysis

CS F-415 Data Mining | Spring 2026
Instructor: Dr. Akanksha Rathore
Weightage: 10% of Course Grade
Group Size: 4 Students

PROJECT OVERVIEW

You are a data mining consultancy hired by a city's **Municipal Corporation**. They want to reduce traffic congestion and optimize public transportation but have no clear plan. They have given you a large dataset of taxi/rideshare trips and said: *"Find something useful."*

Your job is to:

1. **Formulate** rigorous data mining problems from this vague business goal.
2. **Implement** end-to-end mining (preprocessing, modeling, evaluation).
3. **Interpret** your findings as actionable business insights.

Dataset: Real-world taxi trip data (100,000+ records) with timestamps, GPS coordinates, trip duration, and zone IDs. A supplementary weather dataset will be released in Week 9.

LEARNING OBJECTIVES

By completing this project, you will:

- Apply the KDD Process (Data Understanding → Preprocessing → Mining → Interpretation).
- Handle messy real-world data (missing values, outliers, integration challenges).
- Compare multiple algorithms and justify selection based on metrics.
- Translate mathematical patterns into business-readable insights.

PROJECT PHASES & DELIVERABLES

Phase 1: Problem Formulation & Exploratory Data Analysis

Timeline: Weeks 1-8 (Due: March 2, 2026)
Deliverable: 2-page Proposal (PDF)

Tasks:

1. **Data Exploration:**
    - Load and explore the dataset.
    - Report basic statistics (number of records, features, data types).
    - Identify data quality issues (missing values, outliers, inconsistencies).
2. **Problem Formulation:**
    - Define **3 data mining tasks** that address the client's goal.
    - For each task, specify:
        - **Mathematical Task** (Classification/Clustering/Association/Regression/Anomaly Detection).

- **Input/Output** (e.g., "Input: Pickup Time, Zone → Output: Predicted Duration").
- **Business Value** (How does solving this help reduce congestion?).

Phase 2: Implementation & Analysis

Timeline: Weeks 9-15 (Due: April 21**,** 2026)
Deliverable: Final Report (5 pages max, PDF) + Code (Jupyter Notebook)

Tasks:

1. **Data Preprocessing:**
   - Handle missing values, outliers, and duplicates.
   - Feature engineering (e.g., extract "Hour of Day," "Day of Week" from timestamps).
   - **Integration Challenge (Week 9):** A weather dataset (Temperature, Rain) will be released. Integrate it with trip data to analyze weather impact.
2. **Modeling:**
   - Implement algorithms for your 3 tasks (using libraries like scikit-learn is allowed).
   - For at least ONE task, compare two algorithms (e.g., K-Means vs. DBSCAN for clustering).
   - Tune hyperparameters and report evaluation metrics (Accuracy/Silhouette Score/RMSE/Confidence, as applicable).
3. **Interpretation & Business Insights:**
   - Explain patterns in plain English (e.g., "Cluster 1 represents 'Airport Runs' with long distances and high fares").
   - Provide **3 actionable recommendations** (e.g., "Add bus stops in Zone X during 6-8 PM").
4. **Failure Analysis:**
   - Discuss where your model failed or struggled.
   - Explain why (e.g., "Prediction accuracy dropped on public holidays due to sparse data").

REPORT STRUCTURE (Final Submission)

1. Executive Summary (0.5 page)

2. Data Understanding & Preprocessing (1 page)

3. Methodology (1.5 pages)

4. Results & Interpretation (1.5 pages)

5. Limitations & Future Work (0.5 page)

SUBMISSION GUIDELINES

What to Submit (via Course Portal):

1. **Final Report (PDF):** 5 pages max (excluding references). Font: 11pt, Single spacing.

2. **Code (Jupyter Notebook):** Well-commented, runs without errors.
3. **Dataset:** If you cleaned/transformed the original, submit your processed version.
4. **Group Contribution Statement:** One sentence per member describing their role.

Deadline: April 21, 2026, 11:59 PM
Late Penalty: 25% (max 2 days; after that, 0 marks).

ACADEMIC INTEGRITY & AI POLICY

- **Open Book:** You may use textbooks, course slides, and online resources.
- **AI Tools (ChatGPT, Copilot):** Allowed for code debugging and brainstorming, BUT:
  - You must **understand and explain** every line of code.
  - Direct copy-paste of reports will be flagged (we check originality).
  - **Interpretation** and **failure analysis** sections must be original. Generic AI-generated text (e.g., "The model showed good performance") will receive 0 marks.
- **Plagiarism:** Groups copying from each other or online sources will receive 0 for the project and face disciplinary action per BITS rules.